3.1 To keep units consistent, start by converting to micrometers, $x = 100\text{nm} = 0.1\mu\text{m}$.

First, calculating values of D for wet and dry oxygen processes, assuming 100 Si and 1000°C = 1273.15°K, using $D = D_0 e^{\frac{-E_A}{kT}}$

$$D_{\text{wet}} = (9.7 \times 10^7)e^{\frac{2.05}{0.1097}} = 0.7430 \text{ (linear)} = \frac{B}{A}$$
$$D_{\text{wet}} = 386e^{\frac{0.78}{0.1097}} = 0.3152 \text{ (parabolic)} = B$$
$$D_{\text{dry}} = (3.71 \times 10^6)e^{\frac{2.00}{0.1097}} = 0.0448 \text{ (linear)} = \frac{B}{A}$$
$$D_{\text{dry}} = 772e^{\frac{1.23}{0.1097}} = 0.0104 \text{ (parabolic)} = B$$

For wet O_2, we assume an initial oxide layer thickness $X_i = 0$ nm, so that $\tau = 0$. Solving for a final thickness of 0.1µm,

$$0.1 = 0.2121[\{1 + \frac{1.2608}{0.1799}(t)\}^{\frac{1}{2}} - 1]$$
$$t = ([\frac{0.1}{0.2121} + 1]^2 - 1)(\frac{0.1799}{1.2608}) = 0.1663 \text{ hrs} = 9.98 \text{ mins.}$$

For dry O_2, we assume an initial oxide layer thickness $X_i = 25$ nm, so that $\tau = \frac{0.025^2}{0.0104} + \frac{0.025}{0.0448} = 0.6181$. Solving again for a final thickness of 0.1µm,

$$t = ([\frac{0.1}{0.1161} + 1]^2 - 1)(\frac{0.0539}{0.0416}) - 0.6181 = 3.58 \text{ hrs.}$$

In conclusion, at $T = 1000$°C, wet oxidation is preferable for the much faster growth rate up to 100nm.

3.3 Given the differential equation

$$\frac{dX_o}{dt} = \frac{J}{M},$$

where J is the oxidizing flux, given as $J = \frac{DN_0}{(X_o + \frac{D}{k_s})}$, and M is the number of molecules of the oxidizing species incorporated into a unit volume of the resulting oxide, then

$$\frac{dX_o}{dt} = \frac{DN_0/M}{(X_o + D/k_s)}.$$

As shown in the textbook, $A = \frac{2D}{k_s}$ and $B = \frac{2DN_0}{M}$, and so the equation can be reduced to

$$\frac{dX_o}{dt} = \frac{\frac{B}{2}}{X_o + \frac{A}{2}}$$

. Using separation of variables,

$$\frac{B}{2}dt = (X_o + \frac{A}{2})dX_o.$$

Integrating both sides,

$$\int \frac{B}{2} dt = \int (X_o + \frac{A}{2}) dX_o$$
$$= \frac{Bt}{2} = \frac{1}{2}X_o^2 + \frac{AX_o}{2}.$$

Solving for t,

$$t = \frac{X_o^2}{B} + \frac{X_o}{\frac{B}{A}}.$$

3.6 (a) Estimating the growth time using Fig. 3.6, $t \approx 7$ hrs.

(b) First finding B and $\frac{B}{A}$ for 1150°C,

$$D_{\mathrm{wet}} = (9.7 \times 10^7) e^{\frac{2.05}{0.1226}} = 5.3316 \text{ (linear)} = \frac{B}{A}$$
$$D_{mathrmwet} = 386 e^{\frac{0.78}{0.1226}} = 0.6661 \text{ (parabolic)} = B.$$

Using Eq. 3.8,

$$t = \frac{4}{0.6661} + \frac{2}{5.3316} = 6.38 \text{ hrs.}$$

3.13